Statement of Purpose

I aim to develop expertise in AMO physics techniques to contribute to cutting-edge research in quantum science. I hope this one day leads to a research career running a high-impact lab. I have demonstrated my potential for this through my extensive experience working with Brian Odom in the Center for Fundamental Physics at Northwestern, which has also strengthened my desire to continue conducting research at a high level. I am specifically interested in the utilization of tabletop atomic systems for fundamental physics and basic science research. As it is a premier place for this research, a PhD experience at Johns Hopkins University would be indispensable.

Research Overview: I joined Professor Brian Odom's lab my freshman year and have since played a key role in commissioning a new doped cryogenic rare-gas crystal experiment from the ground up to study sources of decoherence. These crystals can host orders of magnitude more atoms or molecules than conventional traps and beams, offering vast improvements in statistical sensitivity and making them a promising platform for quantum sensing and fundamental physics. However, their sensitivity is currently limited by short coherence times (T2*). My work has focused on addressing this limitation from two complementary angles. First, I have **developed a novel, highly collimated and tunable dual-species rubidium-argon beam source** to control crystal growth and enable systematic studies linking growth parameters to crystal structure and coherence. Second, I have simulated **spin-spin interaction-induced inhomogeneous broadening** in these crystals, quantifying the contribution of dipolar interactions to the observed linewidths, and identifying optimal dopant densities. Both efforts are being developed into an honors thesis, and the beam source work into an instrumentation paper.

Before beginning my primary research, however, I completed a temporary internship under Professor Tim Kovachy to develop the expertise needed to build a beam source for the Odom lab.

Research Kovachy Lab: In Professor Kovachy's group, I designed a high-temperature strontium vapor cell under the guidance of a PhD student as a reference to enable Sr laser cooling for their atom-interferometry experiment. I fabricated a custom stainless-steel Sr reservoir with thermal-isolation stages and implemented custom Nichrome-mica heaters to achieve the temperatures required for high optical density (>500°C). Additionally, I set up an ultra-high vacuum (UHV) chamber to perform outgassing tests on the heaters and developed an optical layout for the implementation of absorption spectroscopy. The long hours spent debugging heater failures and iterating through multiple reservoir designs allowed me to first discover my love for building novel systems with complex constraints. The experience gave me a foundation in a variety of skill sets, including UHV, thermal engineering, and optics, which I brought to Professor Odom's new project.

Dual-Species Beam Details: The development of my dual-species Rb-Ar beam source has been my primary project and has cemented my desire to pursue experimental AMO physics. I first began by collaborating with a PhD student to design the primary UHV chamber for crystal growth and analysis, as the need for compatibility with cryogenics, X-ray diffraction, and optical access strongly constrained the geometry and placement of the beam source. Within these constraints, I created a method to transport our dopants and rare-gas roughly half a meter to a 1 cm growth substrate using a nozzle array of 231 microcapillaries, producing a highly collimated, mixed Rb-Ar beam. To control the Rb flux and prevent nozzle clogging, I developed

Nichrome-mica heaters and temperature sensors, while an Ar gas-handling system I implemented allows independent control of Ar flux.

Characterizing the beam required assembling a dedicated UHV chamber and building a rubidium spectroscopy setup (see "apparatus diagram and images" in CV). This involved recommissioning an ECDL, implementing frequency modulation, and aligning the full optical chain, complete with Faraday isolation, mode shaping, collimation, fiber coupling, and a reference cell. By fitting absorption spectra at multiple transverse positions along the beam and analyzing the data in Python, I confirmed a half-angle divergence of $(\alpha)_{1/e} = 0.74 \pm 0.1$ degrees, enabling high-efficiency capture at our growth substrate. At the same time, the controlled Rb and Ar fluxes measured enable us to reach the ratios and growth rates required for our planned crystal studies. During this project, I have written standard operating procedures, helped onboard and train new undergraduate students, and presented my results at a poster session at the 2025 Midwest Cold Atom Workshop, all of which have further prepared me for life as a PhD student. My agency over every stage of this project has shown me that I enjoy and can handle the full lifecycle of a complex experimental apparatus, and I plan to use these skills to develop cutting-edge quantum science experiments in graduate school and beyond.

Broadening Simulation Details: Utilizing my background in applied mathematics, I have developed Monte Carlo simulations to model how dipolar interactions between dopants in these dilute spin ensembles contribute to the decoherence we observe. By comparing simulated linewidths to published experimental data, I estimate the contribution of broadening that can be attributed to spin-spin interactions. This is central to the Odom lab's core science goals, as we aim to uncover and mitigate other broadening mechanisms, such as crystal defects. These results now inform our target dopant densities and give us concrete benchmarks for coherence times as we work to improve crystal quality.

Alongside my research, I have taken graduate courses in Quantum Electronics and Advanced Topics in AMO Physics, and I have also attended Johns Hopkins' two-week summer school on Quantum Sensing and Precision Measurement. These experiences have strengthened my conceptual framework and allow me to fundamentally understand and approach more complex questions in AMO physics.

Outlooks: A PhD from Johns Hopkins University would be invaluable to my aim of conducting high-impact research in quantum science and fundamental physics. Interacting with the AMO faculty during JHU's summer school has strongly supported my decision to pursue a degree there. It was Dave DeMille's lectures where I first truly understood what an eEDM was, where its non-zero value could come from, and why searching for time-reversal violations is such a rich probe of beyond-the-Standard-Model physics. And Professor Chris Overstreet's explanation of how measurements are fundamentally counting and comparing things has firmly stuck with me.

It would be an honor to work with Professor Dave DeMille, but I am also very interested in the work of Professor Chris Overstreet and Professor Alex Sushkov. I would be grateful for the opportunity to pursue my PhD with any of these professors, as they conduct cutting-edge research on the atomic systems I am interested in.